

3B. WAP to simulate the working of a circular queue of integers using an array. Provide the following operations Insert, Delete & Display. The program should print appropriate messages for queue empty and queue overflow conditions.

```
void enqueue () {
    if (front == -1 && rear == -1) {
        front = rear = 0;
        queue[rear] = x;
    }
    else if ((rear+1)%N == front) {
        printf("Queue is full.");
    }
    else {
        rear = (rear+1)%N;
        queue[rear] = x;
    }
}
```

```
void dequeue () {
    if (front == -1 && rear == -1) {
        printf("Queue is empty.");
    }
    else if (front == rear) {
        front = rear = -1;
    }
    else {
        printf("Deleted element = %d", queue[front]);
        front = (front+1)%N;
    }
}
```

```

void display () {
    i = front;
    while (i != rear) {
        printf ("%d", queue[i]);
        i = (i+1) % N;
    }
    printf ("%d", queue[rear]);
}

```

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CODE:

```

#include <stdio.h>
#define N5
int queue[N];
int front = -1;
int rear = -1;

void enqueue () {
    int x;
    printf ("Enter a num:");
    scanf ("%d", &x);
    if (front == -1 && rear == -1) {
        front = rear = 0;
        queue[rear] = x;
    }
    else if ((rear+1) % N == front) {
        printf ("Queue is full\n");
    }
    else {
        rear = (rear+1) % N;
    }
}

```

```

    }
    queue[rear] = 2;
}

```

```

void dequeue() {
    if (front == -1 || rear == -1) {
        printf("Queue is empty\n");
    }
    else if (front == rear) {
        printf("Deleted ele = %d\n", queue[front]);
        front = rear = -1;
    }
    else {
        printf("Deleted ele = %d\n", queue[front]);
        front = (front + 1) % N;
    }
}

```

```

void display() {
    int i;
    if (front == -1 || rear == -1) {
        printf("Queue is empty\n");
    }
    else {
        i = front;
        printf("Queue:\n");
        while (i != rear) {
            printf("%d", queue[i]);
            i = (i + 1) % N;
        }
        printf("%d", queue[rear]);
    }
}

```

int main() {

int ch;

do {

print + ("1) Insert 2) Delete 3) Display");

print + ("Enter your choice:");

scanf ("%d", &ch);

switch (ch) {

case 1: enqueue();

break;

case 2: dequeue();

break;

case 3: display();

break;

default: print + ("Invalid input");

}

while (ch != -1);

}

Output:-

1) Insert 2) Delete 3) Display

Enter your choice: 1

Enter a num: 11

1) Insert 2) Delete 3) Display

Enter your choice: 1

Enter a num: 12

1) Insert 2) Delete 3) Display

Enter your choice: 1

Enter a num: 13

1) Insert 2) Delete 3) Display

Enter your choice: 1

Enter a num: 14

1) Insert 2) Delete 3) Display

Enter your choice : 1

Enter a num: 15.

1) Insert 2) Delete 3) Display

Enter your choice : 2

Deleted ele = 11.

1) Insert 2) Delete 3) Display

Enter your choice: 3

~~Queue :~~

~~12 13 14 15~~

~~11~~
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